



An accurate violence detection framework using unsupervised spatial–temporal action translation network

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Abstract

Automatic human behavior monitoring is essential for surveillance cameras in public and private environments. Violent action is challenging because the available violence dataset is insufficient for deep network training. Also, human behavior contains high intra-class variations and inter-class similarities that make violence detection very challenging. In this paper, we proposed an unsupervised Spatial–Temporal Action Translation (STAT) network to accurately distinguish between behaviors and overcome the insufficient violence data problem. Our framework comprises a person detector, motion feature extractor, STAT network, and output interpretation. The proposed framework performed well in different environments because it detects objects in each frame and removes irrelevant background information. As violent motion pattern changes rapidly with high velocity, temporal features play a crucial role in the recognition, and we use it as the input of the STAT network. The STAT network has been trained with normal behavior data, translating normal motion to the spatial frame. Due to the complicated actions in violent behavior, the STAT network cannot reconstruct the violent frame correctly, and therefore, actions will be categorized by comparing the actual and reconstructed frames and measuring the reconstruction error in the output interpretation part of the framework. The proposed unsupervised framework achieved comparable accuracy and outperformed previous works in terms of generality.

Keywords Generative adversarial network · Unsupervised learning · Action recognition · Violence detection

1 Introduction

In recent years, analyzing human behavior to detect violent actions has been one of the hot topics in the computer vision community. Each year, over 1.6 million people die because of violence, and it is one of the leading causes of death for people aged 15–44 worldwide, which accounts for 14% in men and 7% in women [1]. As a result, Closed Circuit Television Cameras (CCTV) in public and private places such as

houses, banks, schools, and prisons are overgrowing. Traditionally, an operator is needed to monitor the cameras, but it is too costly and not affordable in many cases. Scientists have developed intelligent CCTV cameras to automatically analyze the environments and detect suspicious events in video frames in recent years. These systems help reduce the rate of violent activity and provide more safety for the community. Internet safety is another usage of automatic violence detection to protect users from harmful information. Every day, many videos are uploaded on social media like YouTube, Instagram, and Facebook, and intelligent violence detection approaches are the best way to manage them properly. There are many challenges in violence detection systems, such as inter-class similarities and intra-class variations [2], camera viewpoint [3], background clutter [4], and illumination [5]. Intra-class variation means that the actions of each class do not follow a fixed model, and there are many differences between one person and another. On the other hand, the resemblance between normal and violent samples makes inter-class similarities. Unavoidably, a comprehensive and large-scale dataset must consider all mentioned aspects.

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However, there is not enough real-world violence dataset due to privacy concerns. Therefore, the small violent dataset is one of the most challenging problems in designing a violence detection framework.

Machine learning methods are classified as supervised and unsupervised based on human supervision. Supervised learning involves the human effort to label training data. For instance, in supervised violence detection, normal and violent samples are labeled by humans, and models are trained on both classes. Also, existing violent datasets are too limited and do not include intra-class variations and inter-class similarities, which significantly reduces training performance. Normal and violent samples of Hockey, Movies, and Airlab datasets are shown in Fig. 1. As can be seen, the samples of violent class are very similar to each other and actions mostly included pushing, kicking, and punching. A disadvantage of supervised learning is that model learns some limited patterns available in the training datasets, and in the case of new unseen samples, the model performance will degrade considerably. Also, the annotation of the violent class is video-based, while there are combinations of normal and violent behaviors in it. Therefore, methods with supervised learning cannot provide a generalized accuracy based on the small available datasets. On the contrary, the models learn without human supervision in unsupervised learning. For instance, in unsupervised violence detections, violent actions are not labeled, only normal behaviors are used to model normality, and any deviation is considered abnormal. The model learns patterns of normal behavior effectively and can identify whether a new sample is normal. The advantage of this approach is that models can learn many more samples because there is a wide range of normal action datasets [6, 7], and therefore the model generality will be increased.

Most violence detection methods used supervised learning to classify normal and violent behaviors [8–10]. Although they provide high accuracy in distinguishing between behaviors, a generalized framework is not introduced yet to analyze new and unseen data accurately. These methods learn some limited spatio-temporal patterns of the datasets rather than learning to understand the behaviors comprehensively. For example, in [10], authors reported 100% accuracy for their proposed architecture, but testing it on new unseen datasets showed that the accuracy can be degraded to even 56%, and this demonstrated the weakness of the supervised techniques in the case of insufficient datasets. For this reason, a novel unsupervised Spatial–Temporal Action Translation (STAT) network is proposed in this paper that requires only normal datasets for the training phase. The critical parameter of our method is motion pattern, and there are many differences between normal actions such as walking, running, and even sports with violent activities. Normal actions occur with constant patterns, but there is much randomness in violent behaviors, which makes it possible for high-accuracy

classification. This method is based on adversarial training and can translate temporal features to a spatial domain. At the end of the proposed framework, an interpretation part has been designed to check the reconstruction error between actual and reconstructed frames to classify human behavior.

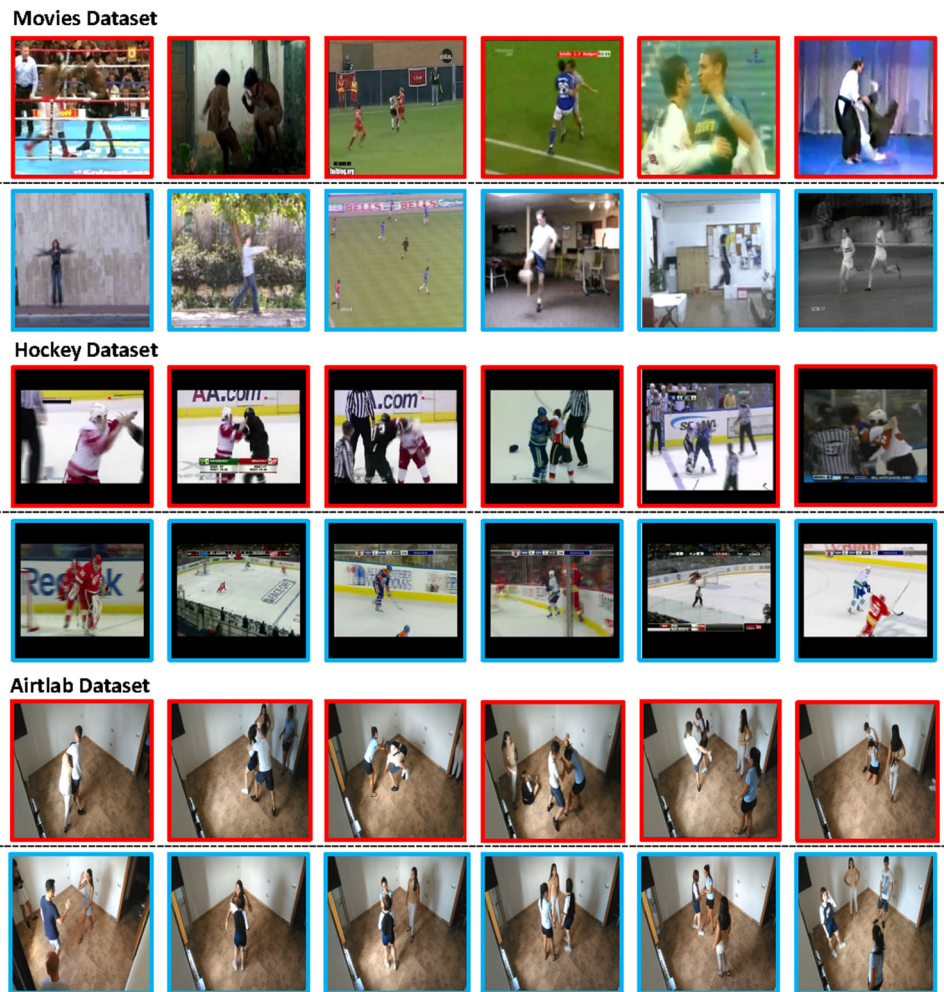
The remainder of this paper is as follows, in Sect. 2, related works of violence detection are reviewed, in Sect. 3, the proposed framework is described in detail, in Sect. 4, experimental results and discussion are presented, and finally, the conclusion is the last section of this paper.

2 Related works

Violence detection is a popular research area in computer vision, and many approaches are presented in this field. They are based on HandCrafted (HC) or deep learning techniques. In HC techniques, violence features are computed manually by human engineering, and different attributes such as motion [11], skeleton [12], trajectories [13], and appearance [14] are analyzed. Violent Flow (ViF) [15] is an HC technique that uses motion information to build discriminative features. Its descriptor has been computed based on the optical flow magnitude distribution variation in consecutive frames over time. To improve the accuracy, Oriented ViF (OViF) is presented in [16]. This technique added the angle of each optical flow vector in the ViF descriptor to consider the feature's entropy. Spatio-Temporal Interest Point (STIP) is another technique in violence detection to extract highly informative regions and remove unimportant background pixels. The author in [17] modeled motions of each STIP neighborhood with the Distribution of Magnitude and Orientation of Local Interest Frame (DiMolif) feature. Another approach based on STIP was introduced in [18]. The distribution of motion in each STIP was modeled with Histogram of Optical Flow (HOF) and Histogram of oriented Gradients (HOG). These histograms are represented with the Bag of Words technique for classification. A trajectory-based violence detection method is proposed in [19]. In this method, each trajectory is obtained using the Kalman filter, and the distribution of optical flow for each of them is calculated. In the end, a support vector machine (SVM) is used for the classification. Some other techniques used appearance information for feature description. In [20], humans are detected by using frame differencing, and the areas, perimeter, and shapes of the bounding box are obtained to extract features. HC techniques cannot provide accurate results for complex actions because they are based on hand engineering and low-level features.

On the other hand, deep learning techniques show better performance without the need for complex hand engineering efforts. In these techniques, high-level features are extracted automatically from each deep neural network (DNN) layer.

Fig. 1 Samples of normal (blue) and violent (red) behavior in different situations of Movies, Hockey, and Airlab datasets



For example, in [21], a 3D convolutional neural network (CNN) is proposed to extract violence features. This network learns complex features as filters convolve in temporal and spatial domains. Using the results of this network, an SVM is utilized to classify normal and violent behaviors. Also, in [22], raw frames are considered as input in a 3D CNN network to learn and classify human behaviors. A combination of motion and appearance features is introduced in [23] to capture spatio-temporal information. This work used two independent CNNs, which take raw frames and optical flow as input. In [24], pre-trained Visual Geometry Group (VGG) network is utilized as a feature extractor. Obtained features are analyzed with Long-Short Term Memory (LSTM) network. In [25], Convolutional LSTM (ConvLSTM) is applied to learn normal activities and detect video anomalies. The advantage of the LSTM network is including temporal dependency among features in the model. Another violence detection approach based on the LSTM network is introduced in [26] that considers acceleration and spatial features as the network's inputs.

Other approaches used a combination of HC and deep learning techniques to improve the accuracy. In [27], a Vi-Net framework is proposed, which used the motion variations as input features and classified the actions with the CNN network. Authors in [28] proposed a pre-processing step based on the Features from Accelerated Segment Test (FAST) detector to extract key point pixels in each frame. The results of the FAST detector are accumulated in consecutive frames to build a representative image. Finally, this image has been given to a 2D CNN network for violence detection. Faster-Region CNN (FRCNN) is applied in [29] to detect objects in each frame. Using the output of the FRCNN network, a Tube CNN (TCNN) network is utilized for action detection. TCNN network has been trained with specific actions such as kicking, punching, hitting, shooting, and throwing. A multi-stream network for violence detection is proposed in [30] to simultaneously consider the attention-based spatial, local spatial, and temporal information. An attention-based stream focuses on some essential regions, including features of action. The temporal stream is based on the optical flow

Table 1 Violence detection approaches using HC techniques

Method	Key point detection technique	Feature extraction technique	Classification technique	Supervision	Scene type
[15, 16]	Not used	Optical flow	SVM	Supervised	Crowded/Uncrowded
[17]	STIP	Distribution of Magnitude and Orientation of Local Interest Frame	BoW	Supervised	Crowded/Uncrowded
[18]	STIP	HOF, HOG	BoW	Supervised	Crowded/Uncrowded
[19]	Gaussian Mixture Model (GMM)	Motion trajectory	SVM	Supervised	Uncrowded
[20]	Frame differencing	Blob appearance information	SVM	Supervised	Uncrowded

Table 2 Violence detection approaches using DNN techniques and combination of HC and DNN techniques

Method	Number of stream	Input	Architecture	Supervision	Scene type
[10]	3	Handcrafted spatial features, motion, frame	2D CNN	Supervised	Crowded/Uncrowded
[21]	1	Sequence of raw frames	3D CNN	Supervised	Crowded/Uncrowded
[22]	1	Sequence of raw frames	3D CNN	Supervised	Crowded/Uncrowded
[23]	2	Raw frames, Optical flow	2D CNN	Supervised	Crowded/Uncrowded
[24]	1	VGG features	LSTM/CNN	Supervised	Crowded/Uncrowded
[25]	1	Sequence of raw frames	ConvLSTM	Supervised	Crowded/Uncrowded
[26]	2	Sequence of raw frames, Acceleration	LSTM	Supervised	Uncrowded
[27]	1	Handcrafted Motion feature	2D CNN	Supervised	Crowded/Uncrowded
[28]	1	Handcrafted Spatial feature	2D CNN	Supervised	Crowded/Uncrowded
[29]	1	Handcrafted Spatial feature	TCNN	Supervised	Crowded/Uncrowded
[30]	3	Handcrafted attention-based spatial feature, Local spatial feature, Motion feature	2D CNN	Supervised	Crowded/Uncrowded
[31]	1	Handcrafted Motion feature	AE	Unsupervised	Crowded/Uncrowded
[32]	2	Handcrafted Motion feature, Person bounding box	AE	Unsupervised	Crowded/Uncrowded

information, which in most previous works has been utilized. The local spatial stream partitions each frame into several blocks for an independent analysis. In multi-stream networks, the final result for the classification is the fusion of each stream. In [31, 32] normal samples are learned with AutoEncoder (AE) networks, and the latent space of AE is analyzed for violence recognition. Framework in [31] is a one-stream temporal AE and in [32] is a double-stream spatio-temporal AE trained to reconstruct the input data at the output. The difference between the embedding of test and normal data is used as a metric for classification. If test data locates far from the embedding of normal data in the latent space, it is classified as violence. Table 1 shows the summary of HC techniques, and Table 2 shows the summary of deep learning techniques and their combination with HC techniques for violence detection. As mentioned in this section, most existing violence detection approaches are supervised, and samples of both normal and violence classes are needed in the training phase. However, they cannot provide high

accuracy in unseen and new data due to the insufficient violence data. Although the proposed frameworks in [31, 32] are unsupervised, they could not achieve significant results regarding the complexity of obtaining distinguishable latent space in AEs. Moreover, compressing the input images to the low-dimension latent space of AEs loses important information, and it could degrade the detection accuracy in detailed frames. In the next section, a generalized framework with a novel unsupervised technique will be proposed To overcome the mentioned problems. The advantage of our framework is adversarial training which could better produce realistic images with a lower reconstruction error.

3 Proposed unsupervised framework

The proposed framework consists of three main parts, person motion extraction, unsupervised Spatial–Temporal Action Translation (STAT) network, and output interpretation. In

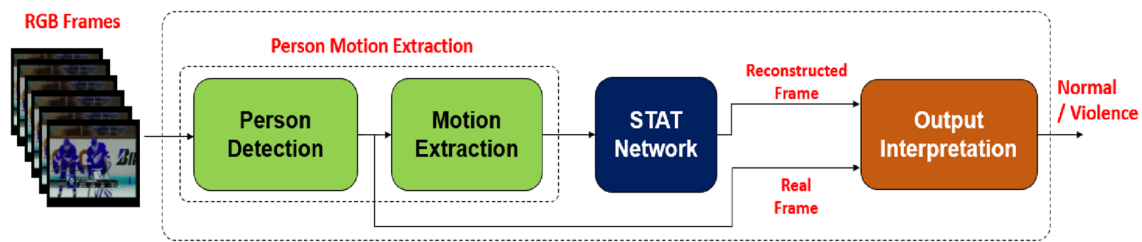


Fig. 2 Block diagram of proposed framework including person motion extraction, STAT network, and output interpretation

Fig. 2, the diagram of this framework is shown. Our framework is object-centric which persons are separated from the other parts of frames. Removing unimportant regions such as background and non-human objects, makes it possible to concentrate just on the body interactions and reduce the effects of environmental conditions. STAT network is trained to translate normal motion to a well-reconstructed body shape. The output interpretation is the final part of the proposed framework, which analyses the STAT network result by comparing the actual and reconstructed frames. As the STAT network is trained on normal patterns, it cannot translate complex violent actions, and by measuring the reconstruction error of actual and translated frames, human behavior can be classified.

3.1 Person motion extraction

In this section, at first, each person's region is detected, and after that, a novel motion descriptor will be performed on each region. Video background contains redundant information such as trees, cars, and houses that are unrelated to the events in the scenes. It is not necessary to train the network with this information. Therefore, we detect humans in each frame and remove all unnecessary backgrounds. There are many unavoidable challenges detecting persons in the frames, such as low quality, person size, and moving camera. Identifying persons in low-resolution videos is difficult, especially when they are far from the camera. Also, moving cameras will change the viewpoint from one frame to another and add complexities to the person detection procedure. Techniques such as background subtraction or frame differencing cannot perform well in these situations. The main weakness of these techniques is background modeling. As persons are detected by subtraction of modeled background and frames, mentioned challenges will degrade the performance. There are some approaches based on deep neural networks that provide accurate results, such as Deformable Parts Model (DPM) [33], Region-CNN (R-CNN) [34], and YOLO [35]. In the DPM method, a sliding window traverses the entire frame to build many sub-frames. Each sub-frame will be analyzed by a deep network for person detection. In R-CNN, a novel regional approach has been proposed to

perform segmentation and obtain candidate bounding boxes. On the other hand, YOLO uses a single regression technique with a single deep neural network to predict bounding boxes and obtain human probabilities. YOLO sees the entire frame at once and is faster than DPM and R-CNN methods. Also, because YOLO was trained on a large-scale dataset, the accuracy and generality of this method are better than others. So, we selected this method for person detection in this paper. YOLO network includes 24 convolutional layers followed by two fully connected layers. Leaky rectified linear and simple linear activation functions are used at the end of hidden and output layers, respectively. The output of this network produces a bounding box with a confidence level of the prediction corresponding to a detected person in the image. The bounding box is created by dividing the entire frame into multiple grids and predicting the confidence level for each grid cell. The confidence level is a probability value that is in the range of [0, 1]. A confidence level at least larger than 0.5 is required to detect a person appropriately. In the YOLO framework, several candidates will be obtained for each frame, and to select a bounding box with the most probability, a non-max suppression has been utilized.

By predicting persons in each frame, the motion patterns of each behavior will be extracted accurately. Motion is a discriminative feature in analyzing human activity. There are many significant differences between normal and violent actions regarding motion magnitude and orientation. Comparing actions like pushing, kicking, and punching with normal actions like walking and talking approves the importance of motion features. Most of the motion features in violence detection methods are based on optical flow extraction. This technique estimates the pixels velocity by comparing two consecutive frames. There are several works proposed for optical flow extraction, such as Lucas-Kanade (LK) [36], Horn-shunk (HS) [37], and Farneback [38]. LK and HS methods used gradient equations to obtain the optical flow with a better computational speed. However, in the case of fast motions, their results contain many errors, and performance degradation is observed. On the other hand, Farneback follows a template-matching approach in consecutive frames. The template is placed around each pixel of the frame, and

optical flow will be calculated by finding the template displacement in the next frame. Farneback shows better results in fast motions than other methods, so we used it in our person motion extraction part. Considering f_x and f_y as template displacement in the direction of x and y , the magnitude of optical flow is calculated as follows:

$$m = \sqrt{f_x^2 + f_y^2} \quad (1)$$

Instead of using raw motion, we used motion variations in our framework. Speed rate is higher in violent actions, and movement orientation changes rapidly. So, to consider this disorder in our descriptor, we compute magnitude and displacement change during time by subtracting consecutive motion attributes. This feature contains three-dimensional vectors in the direction of x and y and magnitude as follows:

$$\text{diff - motion feature} = \{df_x, df_y, dm\} \quad (2)$$

And variations are given by the following equation:

$$df_x = |f_{x_t} - f_{x_{t-1}}|, df_y = |f_{y_t} - f_{y_{t-1}}|, dm = |m_t - m_{t-1}| \quad (3)$$

df_x is motion variation in x dimension, df_y is motion variation in y dimension and dm is magnitude variation. In Sect. 4, we demonstrate that motion variations better discriminate between behaviors than raw motions.

3.2 Unsupervised spatio-temporal action translation network

In this section, we proposed the STAT network to translate temporal aspects of actions to a spatial frame. The input of this network is the motion variation feature which has been obtained in the previous section. This feature will be converted to a spatial image at the output of the STAT network. The STAT network is based on the Generative Adversarial Networks (GANs) [39]. The reason behind choosing GAN as STAT network is its advantages and well-known attributes. GAN is a nonparametric model and is not conformed to data distribution. It automatically finds patterns in the training data and learns to generate new data with the exact statistics of the previously seen data. It implicitly learns data distribution by a min-max game between the generator and discriminator. The main advantage of GAN over similar networks such as AE and Variational AE (VAE) is adversarial training which leads to more realistic images. STAT network trains on normal samples and learns to translate normal motion to the corresponding normal frame. In the testing phase, as the STAT network has been trained only with normal frames, it cannot reconstruct a frame with violent actions, and the output will be a meaningless image. Based on this fact, an

interpretation block will be able to distinguish between normal and violent behaviors. The STAT network is comprised of a generator denoted by G and a discriminator denoted by D . The G part will be trained to map an image x to another image y with the same size ($G : x \rightarrow y$). The D part tries to classify the generated data and label them as real (from the dataset) or fake (generated by the G part). With this adversarial training procedure, real-fake loss for producing realistic images will be obtained as follows:

$$L_{\text{STAT-RF}}(G, D) = E_{x,y} [\log D(x, y)] + E_x [\log (1 - D(x, G(x)))] \quad (4)$$

In this equation, x is the input image including 3 channels of motion feature, y is the actual posture image, and E is the expected value over all real data samples. $G(x)$ is the generated output when real data x is given. $D(x, y)$ is the discriminator's estimate of the probability that real data samples (x, y) is real and $D(x, G(x))$ is the discriminator's estimate of the probability that a generated output is real. The G part of the STAT network tries to maximize the loss function while the D part minimizes it. After reaching a stable point, the training phase of the STAT will be finished. The details of the G and D parts will be described in the following sub-sections.

3.2.1 Generator model

To build the G model, a U-Net architecture proposed in [40] has been constructed. The architecture for the G model is shown in Fig. 3. It can be seen that the architecture is made of an Encoder-Decoder formation with convolutional blocks at each layer. Encoder layers take the input image to down-sample it until the bottleneck layer. There are 8 convolutional layers in the encoder part with demonstrated kernel size and number of filters in Fig. 3. To improve the convergence rate of the network by providing additional paths, skip connections have been inserted in the encoder part to connect layers with the same size. After the bottleneck layer, the decoder part up-sample the result to produce the translated image. For up-sampling, 8 de-convolutional layers are used. Rectified Linear Units (ReLU) is the activation function at the end of each layer which is given as follows:

$$h = \max(0, wx + b) \quad (5)$$

where w is the network weight, b is bias and h is the output of activation function. To prevent overfitting and memorizing in the training procedure, a dropout layer has been added at the end of each de-convolutional layer. G model must generate images similar to the frames in the dataset and for this purpose, a reconstruction loss function for the generator part

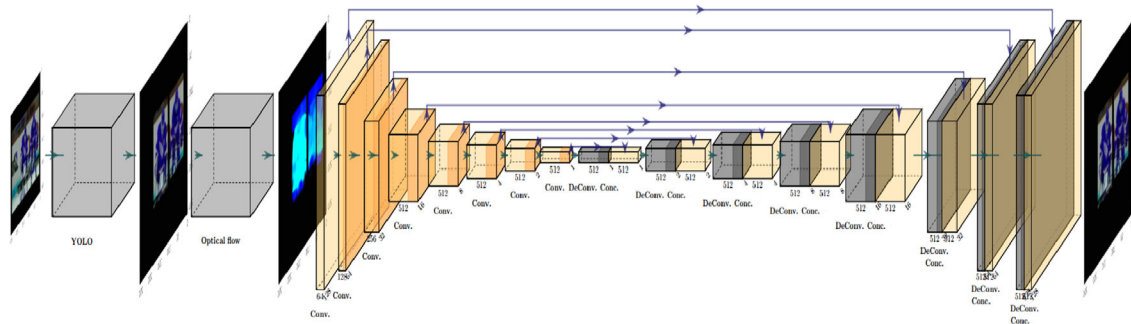


Fig. 3 Generator specifications in the STAT network

is required. The reconstruction loss between the generated posture $G(x)$ and real posture y is computed as follows:

$$L_{\text{STAT-RE}}(G) = E_{x,y}[\|y - G(x)\|_1] \quad (6)$$

$L_{\text{STAT-RE}}$ measures the quality of the reconstructed image. To obtain a well-reconstructed image, the loss value in Eq. (6) should be as low as possible. In this equation, norm $L-1$ is used for comparison, because $L-2$ norm will produce blurry results. Altogether, the optimal model for the generator will be obtained by the following equation:

$$G^* = \arg \min_G \max_D (L_{\text{STAT-RF}}(G, D) + L_{\text{STAT-RE}}(G)) \quad (7)$$

Weights of the STAT network should be updated such that generator losses in the above equation reach the minimum value. In the updating phase, a backpropagation technique is used by computing the gradients of the loss function with respect to the weights. In this paper, the Adam approach is the optimization algorithm for reaching the optimal weights. It is a variant of the gradient descent approach that preserves previous gradients.

3.2.2 Discriminator model

The proposed D model is based on the PatchGAN discriminator. The architecture of this model is shown in Fig. 4. As can be seen, there are 5 convolutional layers for the D model. The input is the concatenation of motion feature and posture image. The posture image can be a real image from the dataset or an image generated by G model. The output shows the likelihood of whether the posture image is a real image from the dataset or it is a fake generated image. A PatchGAN discriminator is different from regular GANs, as there is no fully connected block at the final layer. In regular GANs, an image will be mapped to a single scalar to demonstrate the real or fake output, however, in PatchGAN, an image will be mapped to a matrix and each element of it corresponds to one patch of input. It can be concluded that the classification

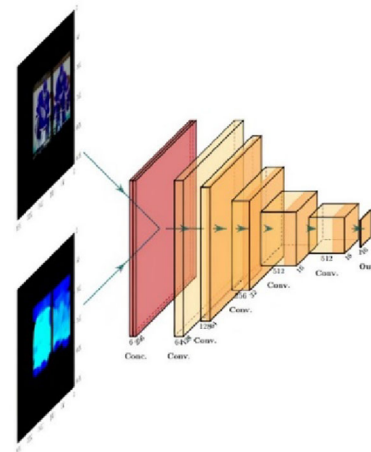


Fig. 4 PatchGAN Discriminator specifications in the STAT network

will be performed for each patch of the input image and the real or fake output is obtained by averaging all patches. So, this model provides better accuracy compared to the regular GANs as it will locally analyze the input image.

3.3 Output interpretation

After the training phase, only the G model is considered in our proposed violence detection framework. For every 3 consecutive frames, persons are detected to obtain their differential motion features. The STAT network takes these features and produces a RGB image. If the features include normal actions, the reconstructed image will be obtained accurately. However, the motion patterns of violent actions are complicated and the output image will be meaningless. To make decisions based on the mentioned reconstructing error, a similarity module shown in Fig. 5 is considered. Image similarity metrics such as PSNR and MSE can be used for analyzing the similarity between the actual and reconstructed frames. As they perform a pixel-by-pixel comparison, they do not perceive human-level abstraction between two images. For high-level interpretation and perception, a pre-trained

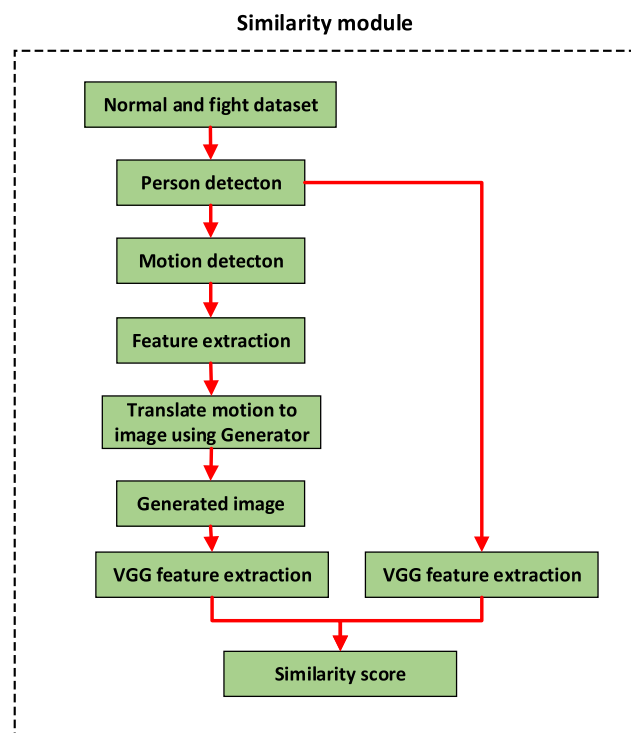


Fig. 5 Flowchart of the proposed output interpretation

VGG network is used based on the procedure of Fig. 6. VGG is a deep CNN trained on over 14 million images of 1000 classes and can extract valuable features from images. We used the VGG network as a feature extractor to convert the output image of the STAT network and actual frames to feature maps. Different layers of VGG were tested to select the

more informative layer. Block4_conv4 produced more accurate results compared to other layers. These feature maps are analyzed with cosine similarity metric as follows:

$$\text{Cosine}(F_{G(x)}, F_y) = \frac{F_{G(x)} \cdot F_y}{\|F_{G(x)}\| \|F_y\|} \quad (8)$$

where $F_{G(x)}$ is the feature vector of generated image and F_y is the feature vector of the actual image. A higher Cosine value shows normal behavior in the frame. To decide whether an action is normal or violent, a threshold is added to the end of the proposed framework. The range of this threshold is dependent on the selected layer of VGG and the effect of this parameter has been investigated in the next section.

4 Experiments and discussion

In this section, the proposed framework is compared with other state-of-the-art violence detection approaches. Experiments have been carried out in the Google Colab environment based on Tensorflow libraries. Optical flow vectors are extracted for every 2 frames and to obtain the differential motion feature, 3 consecutive frames have been selected. We evaluate our framework based on the Hockey and Movies datasets that include various violent and normal activities.

4.1 Datasets

Hockey fight dataset [3] contains hockey players in a stadium that is captured from the hockey games of the National

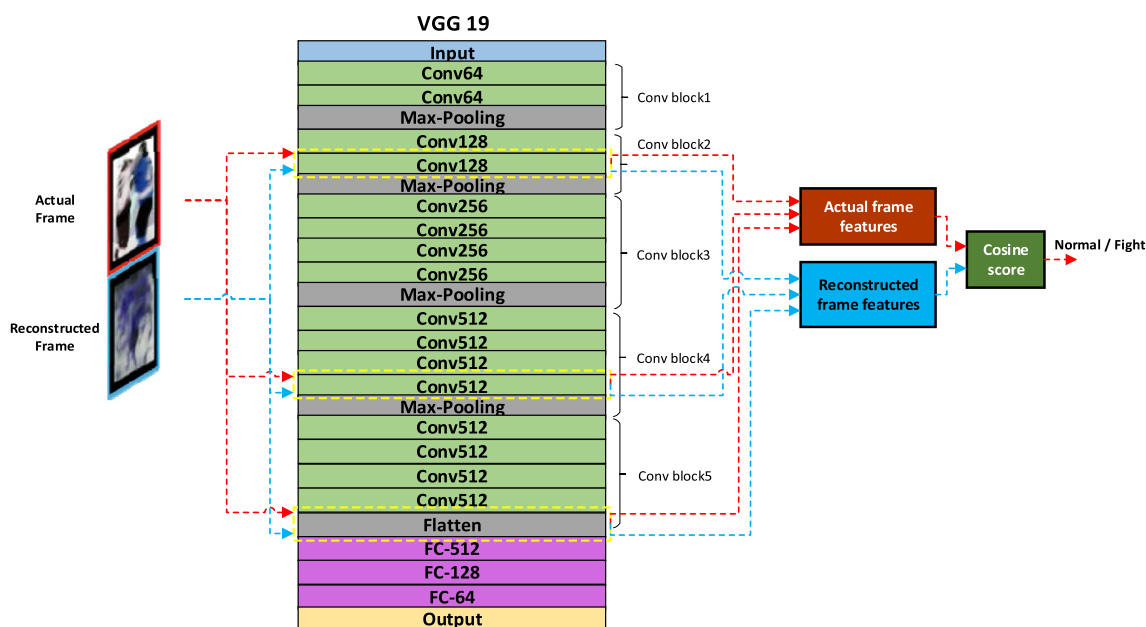


Fig. 6 Output interpretation using VGG architecture

Hockey League (NHL). There are more than 40,000 frames of normal and violent actions in this dataset. Total number of clips is 1000 and half of them are normal and the rest is violent. Each clip consists of 50 frames with 720×576 resolution with a 25 fps. As the camera viewpoint is not fixed and the background changes rapidly, it is a challenging dataset for violence detection approaches.

Movies dataset [3] is gathered from different action movies. It contains 100 clips of normal and 100 clips of violent behaviors. This dataset is less challenging than Hockey dataset because the camera is fixed. The frame resolution is 360×250 pixels, and the frame rate is 25 fps. The duration of the clips varies from 1 to 3 s.

AirtLab dataset [41] consists of two violence and normal classes. Normal class includes different fast-moving actions such as hugging, clapping, cheering, gesturing, and teasing, while violent class includes kicking, punching, slapping, batting, and stabbing. This dataset contains 230 clips for the violent class and 120 clips for the normal class. The duration of the clips varies from 2 to 14 s, with an average length of 6 s. The frame resolution is 1920×1080 pixels, and the frame rate is 30 fps.

4.2 Evaluation metric

For comparison with other methods, Receiver Operating Characteristic curve (ROC) and Area Under Curve (AUC) metric are considered. ROC curve is a graph that illustrate false positive rate and true positive rate. It shows the classifier performance at several thresholds. AUC is another classification metric that measure the overall discrimination of the classifier by considering all thresholds. Also, the performance of the classifier is evaluated with other metrics such as Recall, Selectivity, and $F1$ -score. Recall is the fraction of correctly classified violence samples to the total number of samples in the violence class, and it is formulated as follows:

$$\text{Recall} = \frac{\text{True Positive}}{P} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Negative}} \quad (9)$$

Selectivity is the fraction of correctly classified normal samples to the total number of the normal samples. It is obtained with the following equation:

$$\text{Selectivity} = \frac{\text{True Negative}}{N} = \frac{\text{True Negative}}{\text{True Negative} + \text{False Positive}} \quad (10)$$

Precision is the fraction of correctly classified violence samples to the addition of correctly classified violence sample and normal samples that incorrectly classified to violence

samples, and it computed with the following equation:

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}} \quad (11)$$

$F1$ -score is the harmonic mean of recall and precision, and it is obtained by following equation:

$$F1 - \text{score} = 2 \left(\frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \right) \quad (12)$$

4.3 STAT network setup

75% Percent of all normal actions in the dataset are selected for the training phase, and the rest of them with all violent action clips are considered for the testing phase. To adapt input frame size with our framework, they are resized to 256×256 pixels. The optimizer is Adam with batch size 1 to update the weights of the network for each input sample. The learning rates of the generator and discriminator are set to 0.008 and 0.002, respectively. Figure 7 shows the training results of losses for both D and G models. As can be seen, after about 8000 iterations, the losses of Hockey dataset have been reduced and converged to a stable point. The trained network is capable of reconstructing frames with normal actions correctly. The losses of the Movies dataset also reduced after about 15,000 epochs. We finished the training on both datasets based on the reconstruction error of the training samples. When the network learned to produce realistic images similar to training images, we terminated the training phase.

4.4 Feature map visualization for normal and violent samples

Figure 8 shows the feature maps of encoder and decoder part of the proposed generator. For visualization, one image is randomly selected in each layer. The encoder and decoding parts of normal sample are shown in the first and second row. As can be seen, the convolutional layers compact the optical flow image to a lower dimension, and then, the decoder part accurately generates the frame from the lower dimension. Therefore, each layer of the generator correctly learns the normal motion pattern and the process of converting the motion to the spatial domain. However, in the third and fourth rows, the encoding and decoding parts produce meaningless feature maps, and the final image is incorrect.

4.5 Intermediate qualitative results of each pipeline stage of proposed framework

Figures 9, 10, and 11 show the intermediate qualitative results of each pipeline stage for Movies, Hockey, and Airtlab datasets. The first column shows the real input frames,

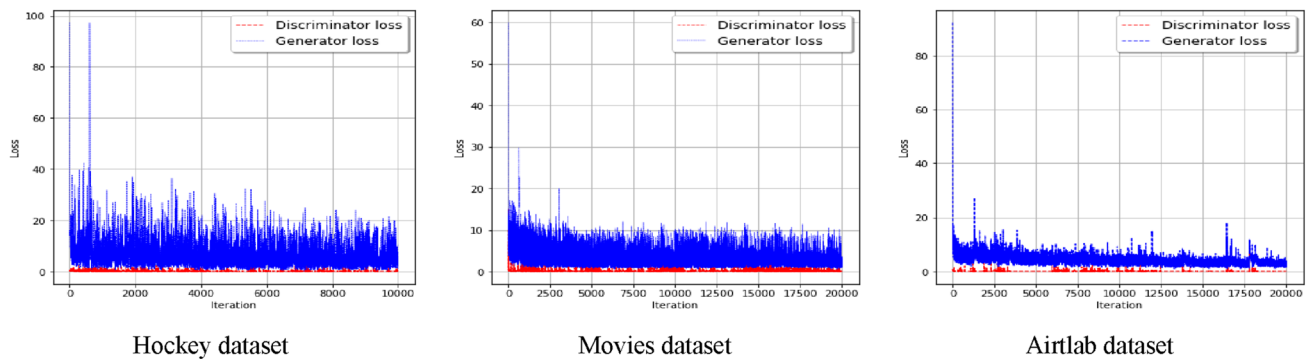


Fig. 7 *D* and *G* model losses in the training phase

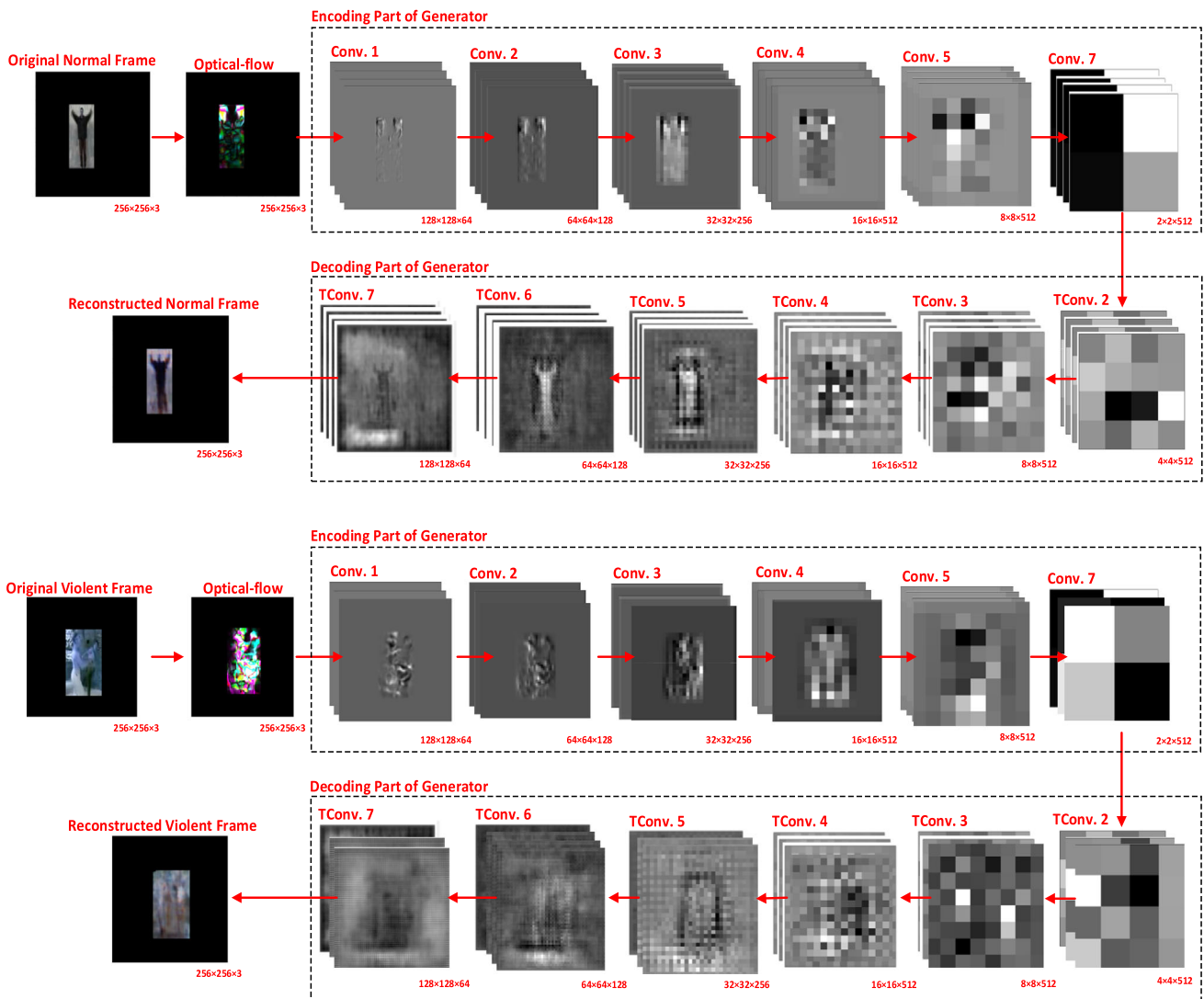


Fig. 8 Visualization of encoder and decoders feature maps for normal and violent samples

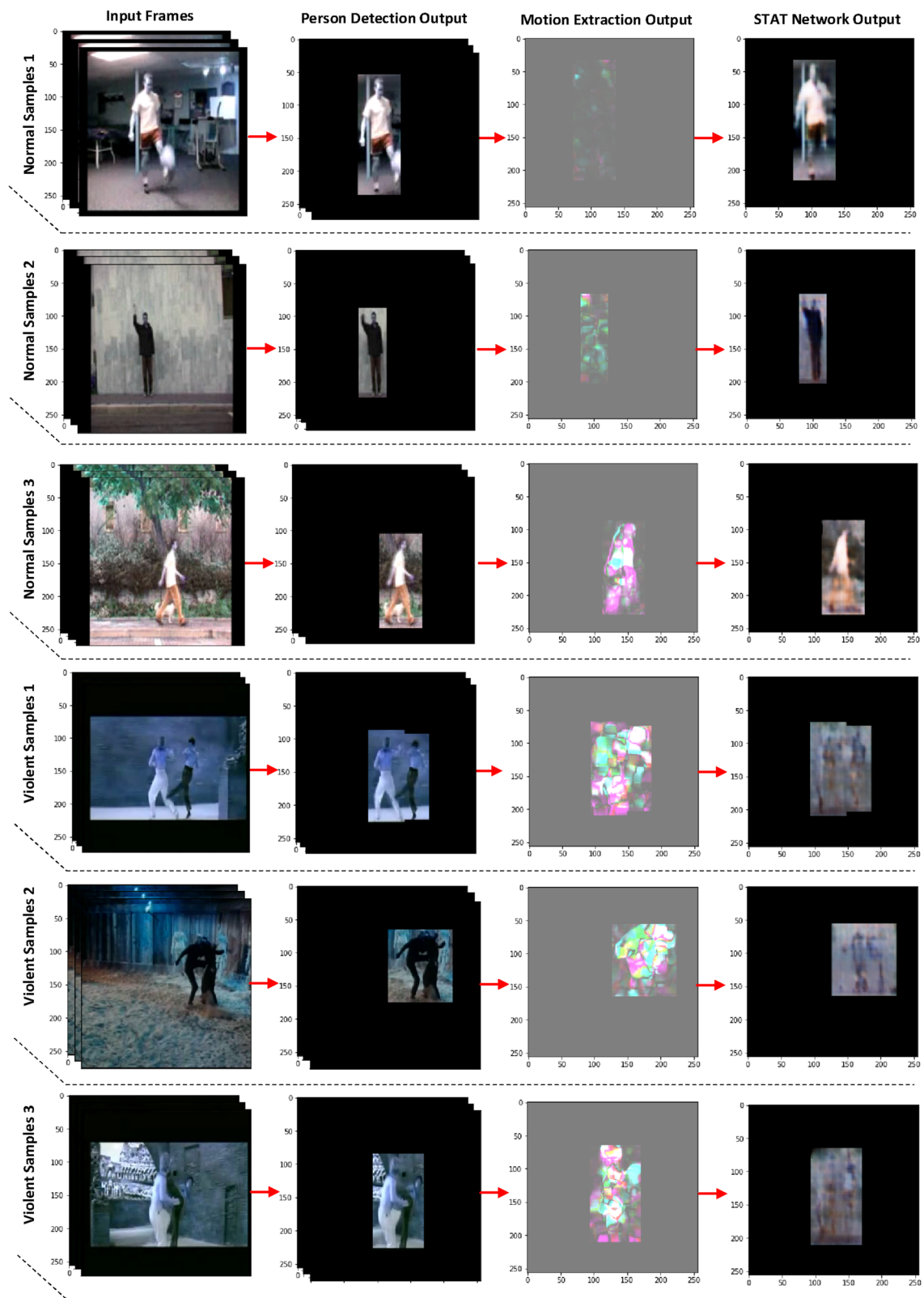


Fig. 9 Intermediate qualitative results of each pipeline stage for the Movies dataset

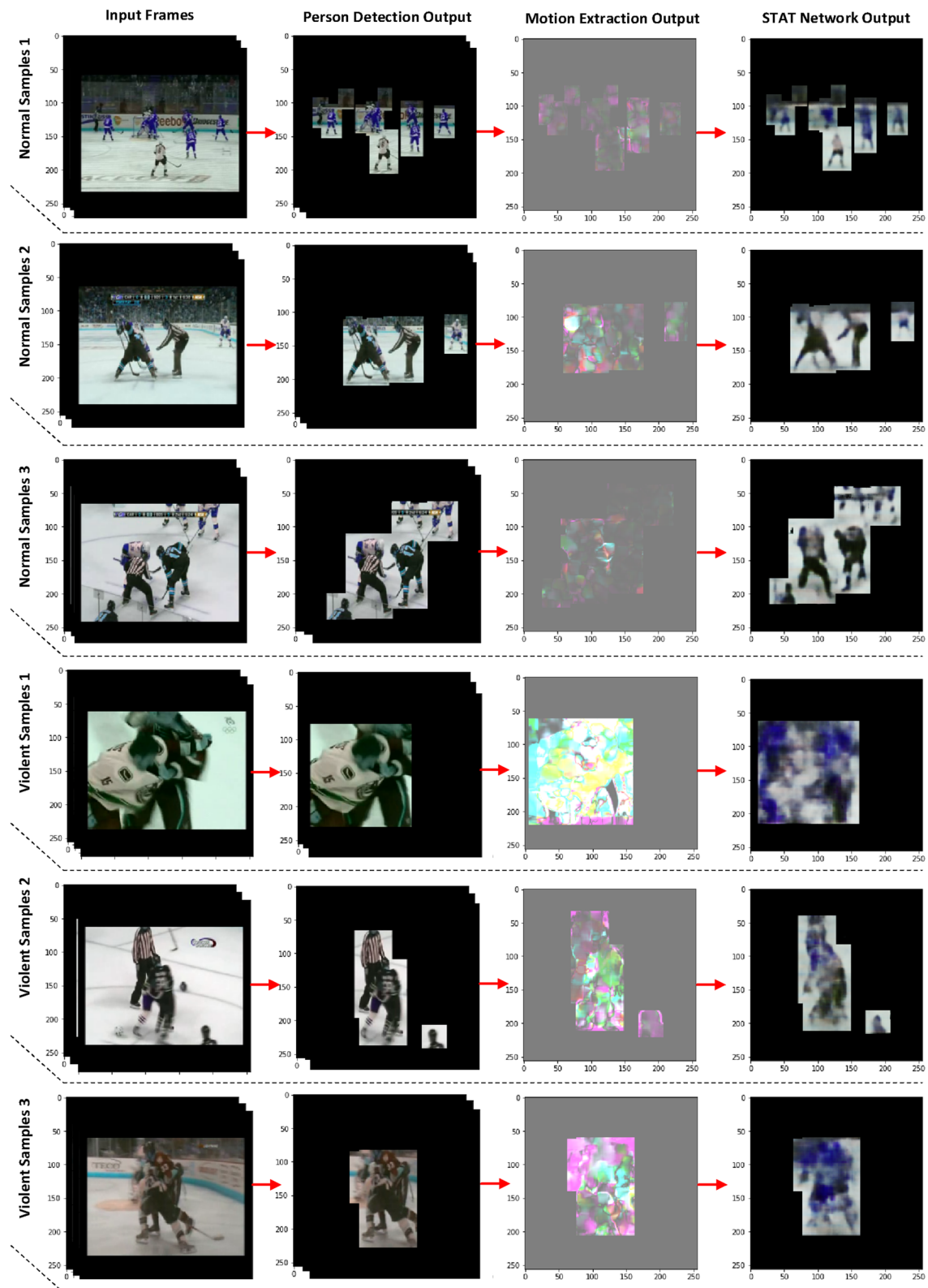


Fig. 10 Intermediate qualitative results of each pipeline stage for the Hockey dataset

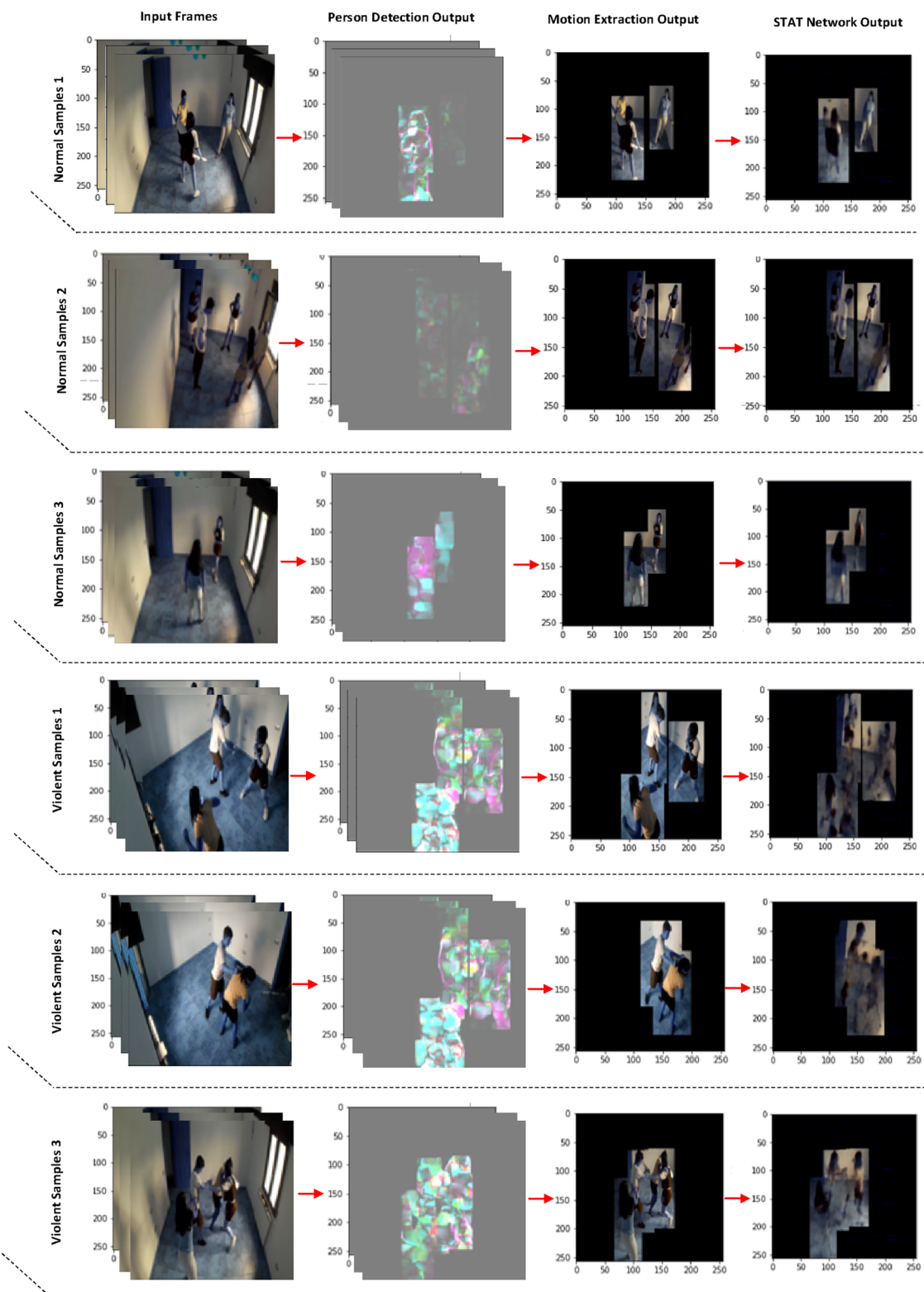


Fig. 11 Intermediate qualitative results of each pipeline stage for the Airlab dataset

the second column shows the person detection output, the third column shows the motion extraction output and the final column shows the STAT output. It can be seen that normal samples in all datasets are reconstructed accurately, while violent samples contain meaningless pixels. Therefore, samples can be classified by computing the reconstruction errors at the output of the STAT network.

4.6 ROC curves for different input features and VGG interpreters

Figure 12 shows the ROC curve of the network based on different inputs and interpreters. We trained the model with magnitude and differential magnitude features to compare features. It can be seen that the differential magnitude feature provides better accuracy in comparison with the magnitude feature and discriminates between normal and violent behavior more accurately. As described in the previous section, the VGG network is used to compute reconstruction error between actual and translated frames. In Fig. 12, we evaluate the ROC result for different layers of VGG in the interpretation part. The first, middle, and final layers, denoted by block2_conv2, block4_conv4, and flatten, are selected for this experiment. The result showed that using the block4_conv4 layer performed better in comparison with other layers. In the first layers of VGG, low-level details such as contours and edges are extracted, whereas, in the last layers, high-level objects are detected. As we are interested in comparing patterns and shapes of hands and legs, the middle layers of VGG are more appropriate. In the remaining sections, differential motion and the middle layer of the VGG interpreter are selected for evaluations. Optimal classification threshold is obtained through the ROC curve and is given by the following equation.

$$Th = \max(\text{true positive} - \text{false positive}) \quad (13)$$

According to this equation, the optimal cut-off threshold maximizes the total true positive samples and minimizes the total false positive samples. For the Hockey, Movie, and Airlab datasets, we selected 0.78, 0.69, and 0.73 cut-off classification threshold, respectively. It means that for the cosine similarity of the reconstructed and real frames higher than the cut-off threshold, the sample is normal and otherwise, it is violence.

4.7 Evaluation results of proposed violence detection framework

Table 3 summarizes the obtained results based on the mentioned evaluation metrics for the proposed STAT network violence detection approach. These results show that the STAT network could classify the normal and violent classes

at an acceptable rate. As the network is trained with normal actions, the performance of the normal class is better than the violent class, and it can be observed from the selectivity metric and recall metric for Hockey and Movies dataset. The *F1*-score, which shows the average harmonic, is obtained 0.936 and 0.991 for Hockey and Movies, respectively, and it clearly approves the high performance of the proposed framework.

4.8 AUC comparison with the state-of-the-art violence detection approaches

In Table 4, an extensive comparison has been carried out with the previous violence detection approaches. The approaches are classified into 3 supervised, semi-supervised, and unsupervised groups. Most previous works proposed for violence detection are based on supervised training. As our work is an unsupervised approach, this fact should be considered for a fair comparison. ViF [15], a supervised motion-based handcrafted approach, has 83% AUC for Hockey and 98% for Movies. DiMolif [17] is another handcrafted method that modeled the motion distribution, and obtained 93% AUC for the Hockey dataset. The method proposed in [18] is based on STIP and Bag of Words (BoW) modeling and provided 86% AUC for Hockey and 82% for Movies. Fast fight [20] used human blob to calculate features and has 76% AUC for Hockey and 97% for Movies datasets. 3D CNN [22] used raw frames and obtained 90% AUC in the Hockey dataset. In [26], authors used a combination of acceleration and spatial features as the input of LSTM and achieved 93% AUC in the Hockey dataset. Another 3D CNN [42] network provides 96% AUC for Hockey and 99% for Movies. HF [28] is a combination of handcrafted and deep methods and has 94% AUC for Hockey and 99% AUC for Movies. 2D CNN [10] showed the highest AUC among the previous works as they used processed data and features for learning. ConvLSTM [41] models the temporal dependency among frames with the LSTM network and achieved 99% AUC for Hockey and 97% AUC for the Airlab dataset, respectively. DLLS [45] is a semi-supervised approach that trained on fewer data compared to supervised methods and obtained 97% AUC for the Hockey dataset. DABA-Net [31] is an unsupervised temporal AE network, and it achieved 82% and 92% AUC for Hockey and Movies, respectively. Double-AE [32] added spatial information to the previous network and obtained 96%, 98%, and 96% for Hockey, Movies, and Airlab, respectively. The results of the proposed framework are obtained 94% and 98% for Hockey and Movies datasets, respectively. Although STAT is trained only on normal samples, the achieved result is acceptable and slightly lower than supervised and semi-supervised approaches. The superiority of our work is that there is no need for hand labeling of violent data, and it saves a lot of time and cost. Another advantage

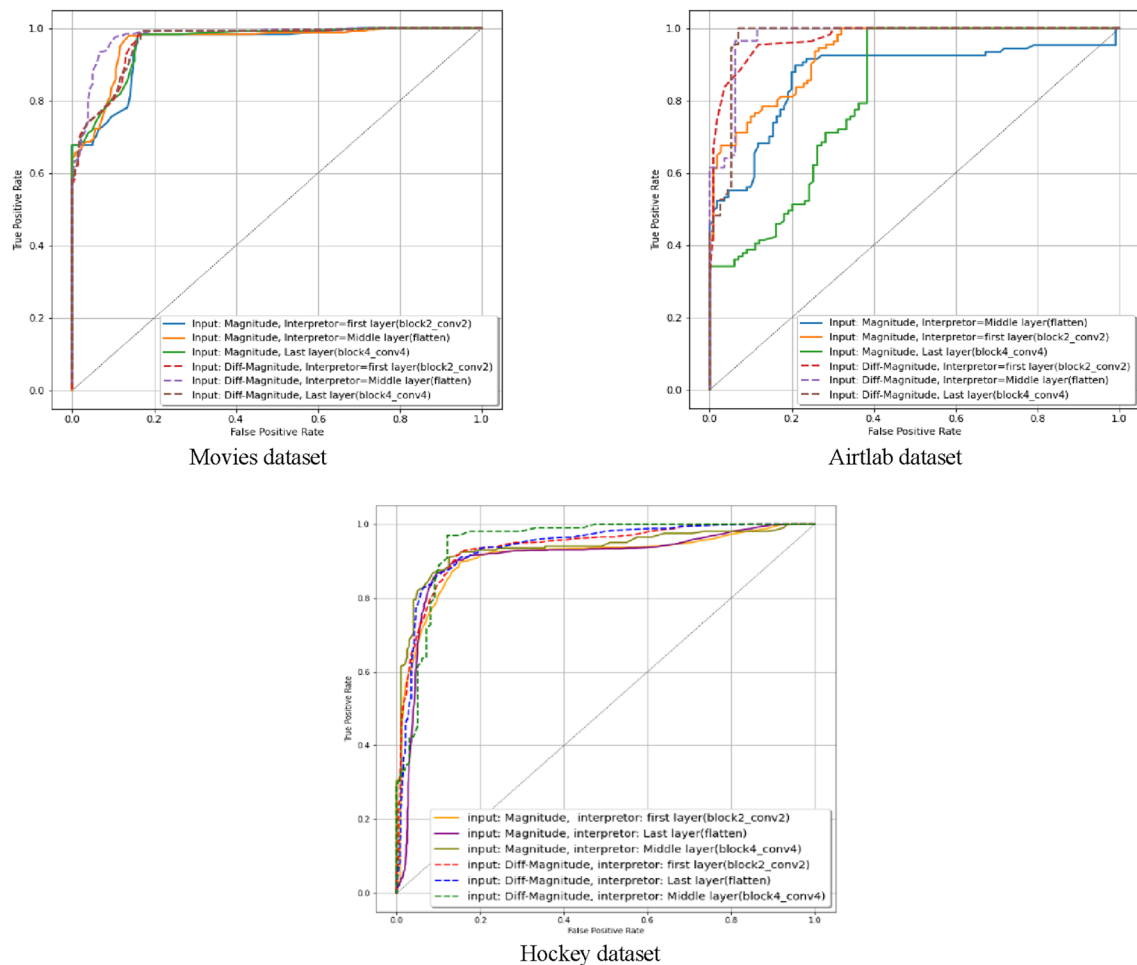


Fig. 12 ROC curve based on different input features, different VGG interpreters, and different datasets

of this work is that it can be trained on large datasets, as only videos with normal actions are required. On the contrary, supervised methods need large violent datasets, including violence activities with high intra-class variations to obtain a high generalization. As a result, although the supervised approaches provide high accuracy for the learned samples, in the case of new samples, their performance will degrade substantially. Also, our work obtained better results compared to other unsupervised violence recognition methods, especially in the Hockey dataset. In [31, 32], autoencoder networks could not reconstruct Hockey output accurately due to the challenging environment and camera motion. However, as our proposed method is based on adversarial training, the discriminator screens the generator's output and helps it to reconstruct accurate results. Therefore, our method could better produce realistic images, and the results are better than autoencoder-based networks.

4.9 Ablation study

In this section, we evaluated our framework without the person detection part to show its effectiveness on the performance of violence detection. Table 5 shows the result of this experiment and the STAT network is trained by considering the entire frame in the sample classification. As can be seen, the AUC metric for all Hockey, Movie, and Airlab datasets has been decreased since irrelevant and redundant information of unimportant regions are added to the framework pipeline. The AUC degradation is 8%, 7%, and 5% for Hockey, Movie, and Airlab datasets, respectively. According to this experiment, it can be claimed that the person detection part in our framework is essential to achieve high accuracy results.

4.9.1 Generality analysis

The most challenging problem in violence recognition is the small size of the violent datasets. Therefore, the model learns few available violent samples, and the performance of the

Table 3 Evaluation results of proposed violence detection framework

		Dataset	Total Positive samples	Total Negative samples	TP	TN	FP	FN	Recall	Selectivity	Precision	F1-score
Unsupervised	Proposed	Movies	3000	450	2768	427	23	232	0.92	0.94	0.991	0.955
		Hockey	20,000	3000	18,251	2793	207	1749	0.91	0.93	0.988	0.949
		Airlab	30,000	3700	23,670	3093	607	6330	0.78	0.83	0.974	0.894
Supervised	Double-AE [32]	Movies	–	–	281	239	49	21	0.93	–	0.851	0.888
		Hockey	–	–	1467	1612	572	449	0.76	–	0.719	0.741
	[24]	Movies	20	20	20	19	1	0	1.00	0.95	0.95	0.975
		Hockey	150	150	148	145	5	2	0.98	0.96	0.967	0.976
	[42]	Movies	50	57	50	57	0	0	1.00	1.00	1.000	1.000
		Hockey	271	241	262	230	11	9	0.96	0.95	0.959	0.963
	[43]	Movies	–	–	–	–	–	–	1.00	1.00	–	–
		Hockey	–	–	–	–	–	–	0.97	0.98	–	–
	[41]	Hockey	–	–	–	–	–	–	0.97	0.97	–	0.978
	[44]	Movies	–	–	–	–	–	–	0.93	0.83	–	–
		Hockey	–	–	–	–	–	–	0.88	0.84	–	–

Table 4 Evaluation results of proposed violence detection framework

Supervision	Method	Hockey (AUC %)	Movies (AUC %)	AirtLab (AUC %)
Supervised	ViF [15]	83	98	N/A
	DiMolif [17]	93	N/A	N/A
	STIP + BoW [18]	86	82	N/A
	Fast Fight [20]	76	97	N/A
	3D CNN [22]	93	87	N/A
	Multi-stream + LSTM [26]	93	N/A	N/A
	ST-3D CNN [42]	96	99	N/A
	HF [28]	94	99	N/A
	2D CNN [10]	100	100	N/A
	ConvLSTM [41]	99	N/A	97
Semi-supervised	DLLS [45]	97	N/A	N/A
Unsupervised	DABA-Net [31]	82	92	N/A
	Double-AE [32]	84	98	N/A
	This work	96	98	94

Table 5 Ablation study results on person detection part of proposed violence detection framework

Dataset	AUC Without person detection (%)	AUC With person detection (%)
Hockey	88	96
Movie	91	98
Airlab	89	94

model in new unseen condition is deficient. To overcome this challenge, we proposed a novel unsupervised framework that trained on only normal samples. Since publicly available normal datasets are large-scale, this framework better learns to discriminate between normal and violent classes compared to supervised approaches. In Table 6, we prove that our model better generalizes to unseen actions by learning the model on a dataset and testing it on other datasets. For this purpose, we trained our network on the Hockey

Table 6 Generality experiment result for new unseen data

Training dataset	Testing dataset	AUC (%)					
		Our method	ST-3D CNN [42]	3D CNN [22]	2D CNN [10]	DABA-Net [31]	Double-AE [32]
Hockey	Movies	97	49	48	56	88	97
	Hockey	96	96	87	100	82	84
	AirtLab	90	N/A	N/A	N/A	N/A	N/A
Movies	Movies	98	99	93	100	92	98
	Hockey	92	63	59	71	73	71
	AirtLab	83	N/A	N/A	N/A	N/A	N/A
AirtLab	Movies	89	N/A	N/A	N/A	N/A	N/A
	Hockey	84	N/A	N/A	N/A	N/A	N/A
	AirtLab	94	N/A	N/A	N/A	N/A	N/A

dataset and tested it on the Movies dataset. Our framework classified samples with a higher AUC than Spatio-Temporal 3D CNN (ST-3D CNN) [42], 3D CNN [22] and 2D CNN [10], DABA-Net [31], and Double-AE [32]. The AUC of our approach is 97%, while ST-3D CNN, 3D CNN, 2D CNN, DABA-Net, and Double-stream AE obtained 49%, 48%, 56%, 88%, and 97%, respectively. Similarly, we trained our network on the Movies and tested it on the Hockey datasets. We achieved an AUC of 92%, while the ST-3D CNN, 3D CNN, 2D CNN, DABA-Net, and Double-stream AE obtained 63%, 59%, 71%, 73%, and 71%, respectively. As can be seen, our approach surpasses all these methods in terms of generalization. Because ST-3D CNN and 3D CNN use raw information as the input of the network, a large-scale dataset is needed to perfectly learn the patterns of human actions. Therefore, they provided lower generality. 2D CNN utilizes the combination of handcrafted information and deep networks and, as a result, provides slightly higher generalization ability than end-to-end approaches. Because DABA-Net used only motion information, it provided lower performance than our method. Double-stream AE is an unsupervised model that embeds temporal and spatial features to lower-dimensional latent space and uses distance metrics as classification. Although it provides better generalization compared to other works, its results are lower than our work. It can be concluded that the unsupervised networks can better capture the patterns in datasets and this improves the semantic learning ability with a label-free approach.

4.9.2 Comparison with anomaly detection approaches

In this section, the proposed violence recognition framework is evaluated on the UCSD pedestrian dataset. This dataset is divided into two parts Ped1 and Ped2. Both parts include crowd pedestrian walkways in which walking people are

annotated as normal class and cars, biking, or skating in walkways are annotated as anomaly class. Since non-person objects are considered anomalies in this dataset, we removed the person detection part from our pipeline. The intermediate qualitative result of each pipeline stage for the UCSD dataset is shown in Fig. 13. As motion patterns of anomaly conditions are different from walking action, they are not reconstructed at the output, and therefore, classification can be performed with reconstruction error. The comparison of the proposed method with the state-of-the-art anomaly detection works is reported in Table 7. As can be seen, our framework could achieve considerable results compared to other works. Since the proposed framework is based on chaotic motion distribution modelling, its capabilities will be limited on datasets with uniform motion patterns such as UCSD. In [46], a double-stream AE is trained to model spatial and temporal normal samples. RGB frame is given to the spatial stream while temporal information is obtained from 4 consecutive inputs with a frame differencing technique. Although this method could achieve 97% AUC, it can only be applied to environments with fixed cameras. Moreover, the results on Ped1 UCSD, which includes more challenging environments, are not reported. Authors in [47] used temporal derivatives for a discriminative manifold representation of normal samples, and 79% and 75% AUC were reported for Ped2 and Ped1, respectively. A spatiotemporal consistency-enhanced network is proposed in [48] to detect abnormal events from videos. This work analyzes spatial and temporal information patterns of previous frames to predict future frames. With the prediction error for the classification metric, 97% and 82.5% AUC were achieved for Ped2 and Ped1, respectively. Clearly, this work is highly dependent on environmental situations, and the AUC is degraded when the environment changes from Ped2 to Ped1. A weakly supervised method is proposed in [49] to detect and localize abnormal behavior recognition.

Fig. 13 Intermediate qualitative results of each pipeline stage for the UCSD dataset

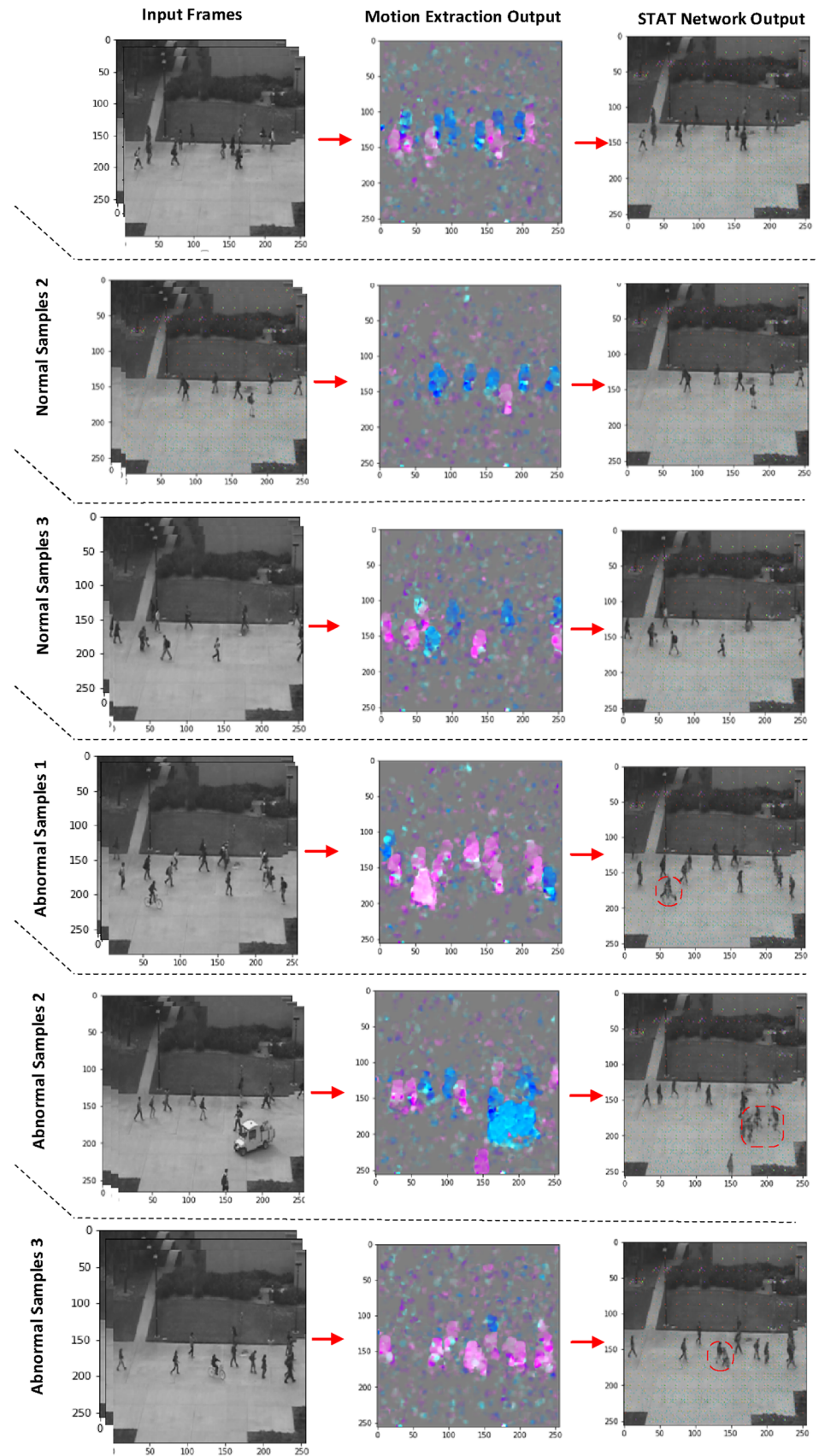


Table 7 Evaluation results of proposed violence detection framework on UCSD dataset

Supervision	Method	AUC% Ped2	AUC% Ped1
Weakly Supervised	HLSOF [49]	80	71
Unsupervised	SVD-GAN [52]	76	73
	DMR [47]	79	75
	SR-VAE[51]	89	90
	R-STAE[50]	83	N/A
	ST-AE [46] (Fixed Camera)	97	N/A
	ST-CE [48]	97	82.5
	This work	93	90

Human actions are represented with the histogram of large-scale optical flow and SVM classified samples with 80% and 71% AUC for Ped2 and Ped1, respectively. A residual spatio-temporal AE is introduced in [50] to model normal events, and about 83% AUC is reported for Ped2. In [51], sparse representations are obtained by training a VAE on normal samples, and the probability score was 89% and 90% AUC for Ped2 and Ped1, respectively. A GAN-based framework is designed in [52] to learn spatio-temporal features from input images. For robust reconstruction, a novel Singular Value Decomposition (SVD) loss function is proposed. The reconstruction error is used for anomaly detection, and 76% and 73% AUC were obtained for Ped2 and Ped1, respectively.

4.9.3 Computational time and resources comparison

In Table 8, we compared the processing time and resources of our method with other state-of-the-art approaches. The processing time corresponds to the required time for temporal to spatial translation and output interpretation. The proposed method requires 0.41 s, which is lower than all methods but slightly higher than Fast Fight [20] and 3D CNN [22]. Fast Fight [20] is a hand-crafted method based on simple frame differencing and blob analysis techniques. 3D CNN [22] used the raw frame as the input of the neural network without any preprocessing step. Therefore, its computational time is lower than all methods. However, due to the simple methodology, its classification accuracy is lower than others. Since we used a person detector and optical flow extractor for temporal information, our computational time is slightly higher yet acceptable for real-time violence detection applications.

Table 8 Processing time of different violence detection approaches

Method	Time (sec)	Resources
ViF[15]	3.27	N/A
DiMolif [17]	2.8	N/A
STIP + BoW [18]	N/A	N/A
Fast Fight [20]	0.42	N/A
3D CNN [22]	0.21	NVIDIA Tesla K40 GPU
Multi-stream + LSTM [26]	N/A	N/A
ST-3D CNN [42]	0.81	GeForce-Titan-X GPU
HF [28]	0.46	N/A
2D CNN[10]	0.49	NVIDIA Tesla K80 GPU
DLLS [45]	N/A	2.8 GHz CPU and 16 GB RAM
DABA-Net [31]	N/A	NVIDIA Tesla K80 GPU
This work	0.43	NVIDIA Tesla K80 GPU

5 Conclusion

A novel unsupervised framework for violence detection was introduced in this paper. Previous approaches were based on supervised learning, which degraded the detection generality as there were insufficient violent datasets. We proposed a spatial–temporal action translation network to reconstruct frames from a discriminative differential motion feature. This network was trained on frames with normal behavior, and it was able to reconstruct only normal frames. The classification was performed by analyzing the similarity between actual and reconstructed frames. The experimental results showed the superiority of this unsupervised framework which was comparable with supervised approaches. The main advantage of this approach was eliminating the need for violent class labeling in the dataset and saving a lot of time and cost. In our future works, we want to add more discriminative features in this framework to improve violence detection accuracy.

Data availability Data available on request from the authors.

Declarations

Conflict of interest Authors certified that they have no conflicts of interest.

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